

Preliminary Evaluation of MeshEcho Confidence-Aware Routing for Airtime-Constrained LoRa Mesh IoT Networks

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Abstract—LoRa mesh networking is attractive for low-power Internet-of-Things (IoT) deployments that need wider coverage than a single hop can provide. However, multi-hop LoRa routing has an inherent reliability-airtime tension: redundant flooding improves reachability but increases channel occupancy, while cached source routes reduce transmissions but can fail when a selected path contains a weak or stale hop. This paper presents a preliminary packet-level evaluation of that tradeoff for small LoRa mesh IoT networks. We build a reproducible simulator with a LoRa time-on-air model, log-distance path loss, log-normal shadowing, half-duplex radios, collision/capture behavior, and probabilistic reception from SNR margin. Using this harness, we compare managed flooding, a source-route cache baseline, and a lightweight confidence-aware routing prototype, MeshEcho, that ranks discovered paths and uses bounded recovery forwarding for route-recovery cases. In the frozen 50-node mixed-traffic matrix, MeshEcho reaches 0.735 ACK-confirmed unicast PDR versus 0.389 for the source-route baseline, with 839.6 s versus 971.5 s of aggregate airtime. A paired cache- lifetime audit shows that matching the source-route baseline’s cache TTL to MeshEcho’s 600 s raises its ACK PDR to 0.519, reducing the native mean gap by 0.138 while leaving a 0.209 ACK-PDR gap in the matched bundle comparison. A separate controlled route-conflict experiment shows that the route-selection gain appears only when the shorter candidate contains a materially weaker hop: with 7.5 dB extra loss on that hop, MeshEcho reaches 0.999 ACK-confirmed PDR versus 0.784 for a shortest-path ablation, whereas null, moderate, and reverse controls produce no gain. The evaluation also exposes important fairness boundaries: fixed repeated pairs favor route caching, the default link setting is link-quality saturated and produces very few multi-hop routes, and adaptive recovery is not budget-matched across all protocol variants. A current-code recovery audit shows that disabling route-miss fallback changes ACK PDR by only $+0.009 \pm 0.018$ in a paired 20-seed replay, but does not make the baseline recovery budgets identical. These results support a bounded preliminary claim rather than universal dominance.

An additional ten-seed, one-shot random-pair probe with timeout retries disabled for Smart-CALM gives MeshEcho 0.413 ACK-confirmed PDR and 97.5 s of airtime, versus 0.456 PDR and 139.5 s for source-route caching. Smart-CALM and its no-confidence control reach 0.489 and 0.488 PDR, respectively, with a paired 95% interval for their difference that includes zero. Thus, the efficiency advantage persists outside the frozen matrix, but a general confidence-ranking advantage is not established.

Index Terms—LoRa mesh, Internet of Things, sensor networks, adaptive routing, airtime efficiency, packet delivery ratio

I. Introduction

LoRa radios are widely used in low-power IoT systems because they provide long communication range at modest device cost and energy consumption [1]. Recent surveys and overview articles show that multihop LoRa/LoRaWAN networking can relax star-topology constraints, but it must manage airtime, link variability, and control overhead carefully [2]–[5]. In many deployments, terrain, indoor attenuation, transmit-power limits, and gateway placement constraints can leave coverage gaps. LoRa mesh networking addresses this by allowing intermediate devices to relay packets across multiple hops for environmental sensing, temporary field monitoring, emergency communication, and community-scale telemetry.

The challenge is that LoRa’s low data rate and long packet time on air make redundant forwarding expensive. A flooding-oriented design can reach many nodes, but every rebroadcast consumes scarce airtime and raises collision pressure. A source-route cache can reduce repeated transmissions, but a cached route is only useful while its weakest hop remains good. This creates a practical design question for LoRa mesh IoT networks:

How much forwarding redundancy should a node spend when path quality is uncertain and airtime is scarce?

Recent protocol studies explore several points in this design space, including distance-vector LoRa mesh routing, reusable mesh libraries, spreading-factor clustering, low-latency multihop forwarding, and real-time industrial LoRa communication [6]–[10]. This paper studies the question as a preliminary routing-system evaluation. We compare two fixed-mode baselines with a lightweight confidence-aware protocol named MeshEcho. MeshEcho is the name of our protocol and is not a fork of Meshtastic or MeshCore. The prototype is intentionally conservative: it does not claim to solve all adaptive LoRa mesh routing problems. Instead, it evaluates whether modest path confidence information can expose a better operating point between managed flooding and pure source-route caching.

The contributions of this paper are:

- We present a reproducible packet-level LoRa mesh simulation harness that evaluates routing policies

under common topology, traffic, PHY, propagation, collision, and airtime assumptions.

- We characterize the reliability-airtime tradeoff between managed flooding and cached source routing in repeated-unicast and mixed IoT traffic.
- We evaluate MeshEcho, a preliminary confidence-aware routing protocol that ranks candidate routes using local link-quality evidence and limits fallback flooding to bounded recovery cases.
- We separate general random-topology results from a controlled route-conflict experiment with null and reverse controls, making the mechanism claim and its fairness boundary explicit.

The scope is deliberately preliminary. We focus on simulation-based evidence and small-scale protocol characterization for ICCT Track 8. The implementation reports the reproducible calm-mesh line as MeshEcho; the later online Smart-CALM controller is treated as an exploratory extension rather than as the main claim of this manuscript. A more complete firmware-oriented controller and broader system validation are left for future work.

II. Background and Motivation

A. LoRa Mesh Routing Tradeoffs

LoRa modulation trades data rate for link budget. For IoT sensing, this is often acceptable because payloads are small and traffic is intermittent. In a mesh, however, each relay repeats the same low-rate packet. A route with several hops therefore consumes multiple packet airtimes, and a flood can multiply this cost across many relays.

Current LoRa mesh systems often operate near one of two routing extremes. A managed-flooding approach rebroadcasts packets with randomized delays and duplicate suppression. This is robust because delivery does not depend on one chosen path, but it can generate many transmissions. A source-route cache first discovers a path and then reuses it for later unicast packets. This reduces redundant forwarding when traffic pairs repeat, but can be brittle if the first discovered path includes a weak hop.

The same tension appears in low-power and lossy network routing more broadly. RPL uses objective functions and routing metrics for constrained networks [11], while ETX-style metrics showed that path selection should consider delivery probability rather than hop count alone [12]. LoRa mesh differs in that airtime is particularly expensive and packets are long enough for collision pressure to dominate at modest traffic levels. This motivates a routing policy that treats redundancy as a limited resource rather than as a fixed protocol behavior.

B. Design Goal

Our design goal is not to maximize reliability at all costs. Instead, we seek a policy that can:

- 1) keep the efficiency benefit of source routing when a path appears sufficiently reliable,

- 2) avoid accepting obviously weak discovered paths when alternatives are available, and
- 3) use bounded fallback forwarding only when route discovery or route confidence indicates uncertainty.

This framing is suitable for IoT and sensor-network settings where traffic is light enough for multi-hop LoRa to be useful, but dense enough that unnecessary rebroadcasts still reduce network capacity.

III. System Model

A. Network and Radio Model

We simulate N LoRa nodes uniformly placed in a square area. Nodes share one half-duplex LoRa channel and may relay packets unless explicitly configured otherwise. The default radio profile uses spreading factor SF9, bandwidth 125 kHz, coding-rate index 1 (4/5), 32-byte payloads, 17 dBm transmit power, and a capture threshold of 6 dB, matching a representative SX126x-class LoRa radio configuration [13].

Packet time on air follows the standard LoRa symbol-time calculation [14]. For spreading factor SF , bandwidth BW , payload length PL , coding-rate index CR , explicit header flag IH , CRC flag CRC , and low-data-rate optimization flag DE , the symbol time is

$$T_{sym} = \frac{2^{SF}}{BW}. \quad (1)$$

The packet time on air is the preamble time plus the payload symbol time:

$$T_{oa} = (N_{pre} + 4.25)T_{sym} + N_{payload}T_{sym}, \quad (2)$$

where $N_{payload}$ is computed using the standard LoRa payload-symbol expression. The simulator accumulates total airtime by adding T_{oa} for each transmitted packet.

Propagation uses log-distance path loss with log-normal shadowing:

$$PL(d) = PL_0 + 10n \log_{10}(d/d_0) + X_\sigma, \quad (3)$$

where n is the path-loss exponent and X_σ is a zero-mean Gaussian shadowing term. The received power is $P_r = P_t - PL(d)$. The receiver noise floor uses thermal noise plus a fixed noise figure. The simulator maps SNR margin to packet reception probability through a logistic function:

$$P_{rx} = \frac{1}{1 + \exp(-k\Delta_{SNR})}, \quad (4)$$

where Δ_{SNR} is the difference between measured SNR and the required LoRa SNR threshold for the configured spreading factor. When transmissions overlap, a receiver applies a capture rule: the desired packet can survive only if it is at least the capture threshold stronger than the strongest interferer. For tractability, all packet kinds use the same configured payload length in the current simulator. Source-route header growth and protocol-specific control-frame layouts are therefore not separately charged in the airtime model.

B. Traffic Model and Metrics

We evaluate repeated unicast traffic, mixed traffic, and a separate random-pair probe. Repeated unicast uses a fixed set of source-destination pairs so that route caching has a meaningful opportunity to help. Mixed traffic contains both unicast and broadcast flows, which is closer to many IoT mesh deployments that combine telemetry, control, and local announcements. The random-pair probe draws a new unicast source-destination pair for each application flow and is used to test whether the frozen result depends on pair reuse.

The main metrics are unicast packet delivery ratio (PDR), broadcast coverage, total transmissions, total airtime, airtime per delivery, and collision failures. The frozen main matrix reports means over 20 random seeds, while the random-pair probe reports means over 10 seeds as a fairness diagnostic. Each seed changes node placement, traffic timing, selected pairs, the static per-link shadowing realization, and stochastic reception outcomes. The frozen main matrix also uses the historical shared random stream in which forwarding jitter and reception draws can consume the same generator. The released simulator provides an opt-in independent channel-randomness mode for new runs; this separates reception draws from forwarding jitter and learning exploration, but it still does not create event-by-event common random numbers when protocols generate different numbers of transmissions. In the main matrix, `unicast_pdr` is the source-side ACK-confirmed delivery ratio; destination DATA arrival is reported separately because destination reception does not guarantee a successful ACK return. The main tables use the frozen three-protocol matrix (managed flooding, source-route cache, and calm-mesh); the later Smart-CALM ablation files are retained as historical artifacts and are not used for the quantitative claims in this paper. The comparison is a common-harness comparison, not a claim that every protocol has identical internal recovery semantics: MeshEcho has bounded fallback behavior, whereas the two fixed baselines do not have an identical recovery controller.

IV. Routing Policies

A. Managed Flooding Baseline

The managed-flooding baseline models a Meshtastic-like forwarding behavior at a protocol level [15]. A source transmits a data packet. Relay-capable nodes that hear a new packet schedule a delayed rebroadcast. If a relay hears another copy of the same packet before its scheduled transmission, it suppresses its own forward. Packets stop when the destination receives them or when the hop limit expires. This baseline represents a high-redundancy operating point.

B. Source-Route Cache Baseline

The source-route baseline models a MeshCore-like path-cache behavior [16]. If a source has no cached route to

a destination, it floods a route request (RREQ). The destination returns a route reply (RREP) along the reverse discovered path. The source then caches the path and later unicast packets carry the complete source route. Only the next hop on the source route forwards the packet. This baseline represents a low-redundancy operating point.

C. MeshEcho Confidence-Aware Prototype

MeshEcho keeps the same basic route-discovery structure but changes how route candidates are admitted. During a short discovery window, the destination may observe multiple RREQ paths. Instead of immediately accepting the first path, the prototype scores candidate paths using simple local evidence from received packets. At a high level, the score increases with stronger SNR margin and decreases with collision evidence and additional hops. The selected route is cached with a finite lifetime.

The prototype also includes bounded fallback forwarding. If no route is found, the source sends a fallback flood to recover reachability. The implementation also supports a confidence-triggered delayed fallback behind a source-routed packet, which is canceled if the original delivery succeeds. In the frozen calm-mesh matrix, the confidence threshold is set to zero, so the reported CALM fallback counts are driven by route-recovery cases rather than by an independently tuned confidence-triggered flood. Fallback is therefore a small-radius recovery tool, not a permanent flooding mode.

This section intentionally describes only the preliminary MeshEcho mechanism evaluated in this paper. More aggressive adaptive recovery and implementation-specific control policies are outside the scope of this submission.

V. Experimental Evaluation

A. Setup

Unless stated otherwise, simulations use 50 nodes in a $3000\text{ m} \times 3000\text{ m}$ square area, 1200 s duration, eight repeated unicast pairs, maximum hop count 7, SF9, 125 kHz bandwidth, 32-byte packets, 17 dBm transmit power, path-loss exponent 2.7, shadowing standard deviation 4 dB, and 20 random seeds. Mixed traffic uses a mean offered load of six application packets per minute. The high-shadowing stress case increases the shadowing standard deviation to 6 dB. These settings define a reproducible common-harness comparison, but they are not intended to represent every deployment density or radio range. The repeated-pair workload is deliberately retained as a route-cache efficiency case; it is not a neutral pair-churn workload. The reported tables retain this frozen legacy matrix for traceability rather than silently replacing it with a harder post-hoc parameter choice.

We performed a post-hoc calibration audit without replacing the frozen results. Across 24,500 unordered node pairs, the minimum calculated direct-link reception probability was 0.999909, with no pair below 0.99. The fixed-pair minimum was 0.9999996 and remained 0.9999866 after an

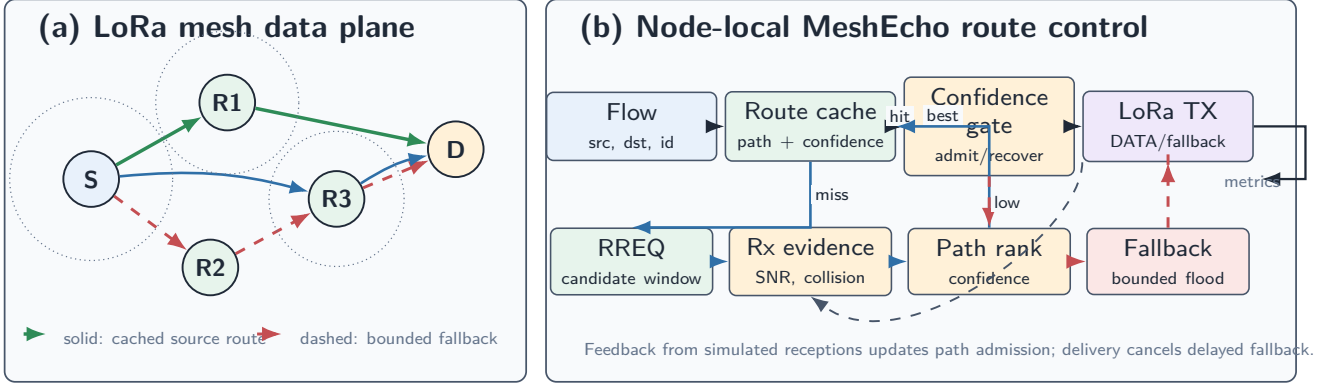


Fig. 1. Implemented MeshEcho routing architecture. Route discovery collects candidate paths, receive evidence is converted to path confidence, and cached source routes are supplemented only by bounded fallback forwarding when path confidence or route availability is insufficient.

additional 8 dB penalty. In the replayed main runs, 128 of 129 MeshCore cached routes and all 142 CALM cached routes were one hop. The matrix is therefore dominated by contention and control traffic rather than multi-hop link-quality conflicts. The static per-link shadowing realization is shared across protocols for each seed, but temporal fading is not modeled. New fairness-oriented reruns should enable the simulator’s `-independent-rng-streams` option, while the frozen files remain in legacy mode for exact reproduction of the published numbers.

B. Random-Pair Non-Saturated Multi-Hop Probe

To test whether the preceding result depends on fixed-pair reuse and link-quality saturation, we ran a separate ten-seed probe with 50 nodes in an 18 km $\times$ 18 km square. The probe uses random source-destination pairs, mixed traffic at four application flows per minute, SF7, 17 dBm transmit power, path-loss exponent 2.75, 4 dB shadowing, and the independent channel-reception random stream. It is a fairness diagnostic rather than a replacement for the frozen 20-seed matrix.

The mean per-seed direct-link PRR quantiles are 0.000026, 0.017577, 0.756816, 0.999517, and 1.000000 at the 5th, 25th, 50th, 75th, and 95th percentiles, respectively. Fractions of direct links below 0.99 and 0.50 PRR are 0.642 and 0.443. The distribution is therefore non-degenerate, although some flows still fail route discovery and the setting should not be treated as a calibrated deployment model.

The Smart-CALM rows in Table I use zero additional timeout retries. This one-shot setting matches the timeout-retry budget across the Smart-CALM variants, while retaining their declared routing and fallback semantics. It is therefore a mechanism-control line, not a claim that all protocols have identical native recovery controllers.

MeshEcho reduces airtime by approximately 30.1% relative to source-route caching in this probe, but its

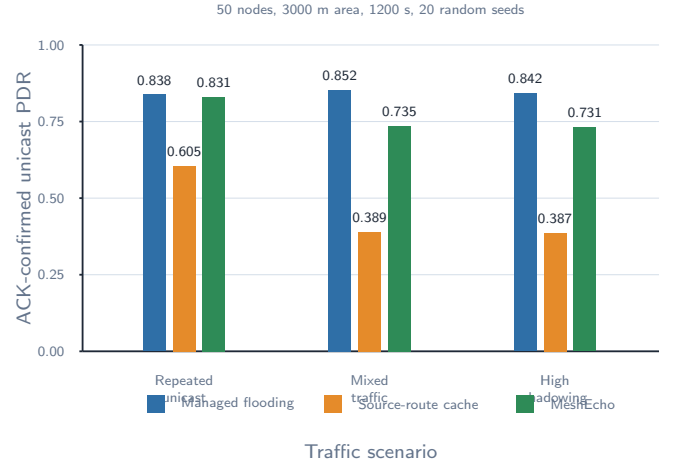


Fig. 2. Unicast delivery ratio across the three evaluation scenarios.

ACK PDR is 0.413 versus 0.456. Under the one-shot budget, Smart-CALM reaches 0.489 ACK PDR and its no-confidence control reaches 0.488. Paired by seed, the full-versus-no-confidence difference is +0.001 with a 95% confidence interval of $[-0.105, 0.108]$; the full-versus-no-fallback difference is +0.024 with $[-0.041, 0.090]$. Both intervals include zero. These results preserve the airtime tradeoff but do not support an unconditional reliability or confidence-ranking claim.

C. Repeated Unicast

Table II and Figs. 2–3 show the repeated-unicast result. Managed flooding reaches the highest unicast PDR in this scenario at 0.838, but spends 1152 s of aggregate airtime and produces 120,276 collision failures. Source-route caching reduces airtime to 358 s, but its PDR drops to 0.605. MeshEcho reaches 0.831 PDR with 256 s of airtime and 17,619 collision failures. Thus, MeshEcho is close to flooding reliability in this particular repeated-pair setting while retaining most of the source-route efficiency.

TABLE I

One-shot, budget-matched random-pair, non-saturated multi-hop probe over 10 seeds. Smart-CALM timeout retries are disabled. Values are means $\pm$ half-widths of two-sided 95% confidence intervals. The multi-hop-route fraction is measured among completed cached routes; it is not defined for managed flooding.

Protocol	ACK PDR	Destination PDR	Airtime (s)	P95 ACK delay (s)	Multi-hop route fraction
Managed flooding	0.680 ± 0.055	1.000 ± 0.000	37.3 ± 4.0	1.67 ± 0.62	–
Source-route cache	0.456 ± 0.074	0.486 ± 0.065	139.5 ± 14.1	0.56 ± 0.41	0.102
MeshEcho	0.413 ± 0.117	0.479 ± 0.108	97.5 ± 10.2	2.49 ± 0.08	0.434
Smart-CALM	0.489 ± 0.122	0.557 ± 0.132	98.7 ± 10.3	3.07 ± 0.43	0.449
Smart-CALM-no-confidence	0.488 ± 0.090	0.566 ± 0.096	97.5 ± 10.3	2.97 ± 0.38	0.328

TABLE II

Repeated-unicast results over 20 seeds. MeshEcho denotes the confidence-aware prototype.

Protocol	Unicast PDR	Total airtime (s)	Collision failures	Airtime reduction vs. flooding
Managed flooding	0.838	1152	120276	–
Source-route cache	0.605	358	27488	68.9%
MeshEcho	0.831	256	17619	77.7%

TABLE III

Mixed-traffic and high-shadowing results over 20 seeds. MeshEcho denotes the confidence-aware prototype.

Scenario	Protocol	Unicast PDR	Broadcast coverage	Total airtime (s)	Collision failures
Mixed traffic	Managed flooding	0.852	0.947	1183	123501
Mixed traffic	Source-route cache	0.389	0.960	971	82454
Mixed traffic	MeshEcho	0.735	0.968	840	74103
High shadowing	Managed flooding	0.842	0.952	1175	123322
High shadowing	Source-route cache	0.387	0.962	970	82324
High shadowing	MeshEcho	0.731	0.965	834	74129



Fig. 3. Aggregate LoRa airtime across the three evaluation scenarios.

Because nearly all cached paths in this matrix are one hop, this result should not be interpreted as a general multi-hop route-selection result.

This scenario favors route caching because the same source-destination pairs repeat. It is therefore a useful cache-efficiency case, but not a neutral workload for protocols that do not rely on route reuse. The result should

be read as a tradeoff point rather than as evidence that MeshEcho dominates flooding or generalizes to arbitrary pair churn. The separate random-pair probe below reports pair churn together with a non-degenerate direct-link PRR distribution and cached-route hop statistics.

D. Mixed Traffic

Mixed traffic is more challenging because broadcast and unicast flows share the same channel. Table III shows that the source-route baseline maintains high broadcast coverage but has a lower ACK-confirmed unicast PDR of 0.389, compared with 0.852 for managed flooding. MeshEcho recovers part of this loss, reaching 0.735 while reducing total airtime from 971 s to 840 s relative to the source-route baseline. Collision failures decrease from 82,454 to 74,103.

Compared with managed flooding, MeshEcho reduces airtime by 29.0% and collision failures by 40.0%, while keeping broadcast coverage within the same range. This end-to-end gain combines route admission and bounded recovery; it should not be attributed to confidence ranking alone. The remaining PDR gap to managed flooding indicates that the preliminary policy spends less redundancy but does not fully replace flood diversity.

E. Recovery-Budget Sensitivity

The frozen main matrix compares the CALM/MeshEcho bundle with a fixed source-route baseline that has no equivalent route-miss recovery controller. To quantify the contribution of this asymmetry, we reran the current code on the same 20 seeds, topology-generation rule, traffic trace, and mixed-traffic parameters while enabling the independent channel-reception stream. The only change between the two CALM variants was whether a failed route discovery could invoke route-miss fallback.

With fallback enabled, CALM reached 0.728 ACK PDR, 0.766 destination PDR, and 847.3 s of airtime. Disabling only route-miss fallback changed these values to 0.719, 0.744, and 831.4 s, respectively. The paired ACK-PDR difference was $+0.009 \pm 0.018$ (two-sided 95% t interval), while the paired destination-PDR difference was $+0.023 \pm 0.023$. Fallback added 62.4 forwarding transmissions and 15.9 s of airtime on average. Thus, route-miss recovery contributes to the end-to-end result, but its mean ACK-PDR effect is small relative to the full CALM-versus-MeshCore-like gap and is not statistically separated from zero in this replay. The corresponding mean route-discovery success rates were 0.798 with fallback, 0.805 without fallback, and 0.834 for MeshCore-like. Thus, the headline CALM result is not explained by a higher route-discovery success rate in this audit, although the native recovery controllers are still not fully budget-matched. The audit is a sensitivity check rather than a fully budget-matched baseline comparison.

F. Route-Cache Lifetime Sensitivity

The native configurations also use different route-cache lifetimes: 300 s for MeshCore-like and 600 s for CALM/MeshEcho. Because the main workload reuses eight fixed pairs, this difference can affect both route-discovery overhead and the number of packets sent on an already discovered path. We therefore reran the same 20 seeds with MeshCore-like at both 300 s and 600 s, while keeping MeshEcho at 600 s.

The native 300 s MeshCore-like setting reached 0.381 ACK PDR, 0.652 destination PDR, and 970.9 s of airtime. Matching its cache lifetime to 600 s increased these values to 0.519, 0.722, and 905.5 s. The paired change was $+0.138 \pm 0.031$ ACK PDR, $+0.070 \pm 0.034$ destination PDR, and -65.4 ± 14.1 s of airtime. Under the matched 600 s TTL, MeshEcho reached 0.728 ACK PDR and 847.3 s of airtime, giving a remaining paired difference of $+0.209 \pm 0.071$ ACK PDR and -58.2 ± 19.2 s of airtime. Thus, the native headline comparison was materially favorable to MeshEcho because of the cache TTL mismatch, although the matched audit still shows a protocol-bundle difference. The frozen native matrix is retained for traceability; the matched audit is the more appropriate fairness reference.

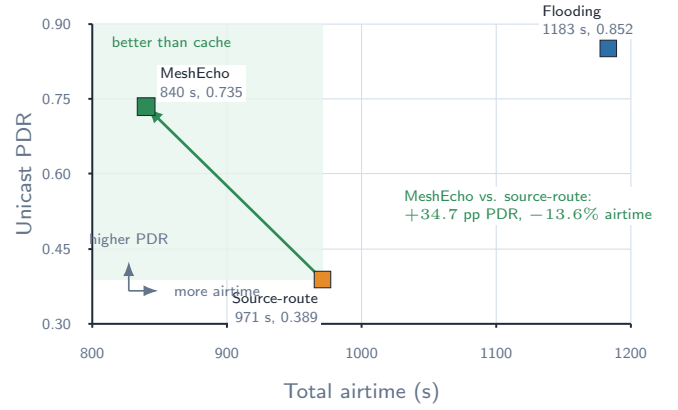


Fig. 4. Mixed-traffic reliability-airtime tradeoff under the evaluated protocol budgets. MeshEcho moves the source-route operating point toward higher delivery with lower airtime, but the end-to-end difference includes bounded recovery not available to the fixed baselines.

G. High-Shadowing Stress Case

The high-shadowing case increases log-normal shadowing from 4 dB to 6 dB. The same qualitative pattern remains, but the absolute values are not identical to the mixed case. Source-route caching reaches 0.387 unicast PDR, while MeshEcho reaches 0.731. Total airtime decreases from 970 s to 834 s and collision failures decrease from 82,324 to 74,129. This shows robustness of the aggregate configuration under stronger shadowing, but does not isolate confidence-aware admission because the random matrix still rarely creates a meaningful length-versus-quality conflict.

H. Controlled Route-Conflict Ablation

The random-topology matrix does not reliably create a direct conflict between shortest-path and confidence ranking, so we use a separate controlled experiment to test that mechanism. The topology has two candidate routes from node 0 to node 5: a short path 0-1-2-5 and a longer path 0-3-4-6-5. Non-edges are blocked explicitly. Each run uses SF9, zero stochastic shadowing, a 12 s discovery window, a 1200 s route-cache TTL, 24 unicast flows spaced by 30 s, and one route-discovery attempt. Online learning, timeout retry, and route-miss fallback are disabled. The confidence variant selects the highest-confidence candidate; the no-confidence variant selects the shortest candidate.

The main mechanism condition adds 7.5 dB path loss to link 1-2, while all other permitted links receive no additional loss. The fairness audit also contains a null condition, a moderate 5 dB short-branch condition, and a reverse condition that weakens link 4-6 on the long branch. Results are paired by seed and use ACK-confirmed PDR.

Table IV shows that MeshEcho does not create a gain when the two branches are effectively equivalent, and it does not blindly prefer the longer route when the long

TABLE IV

Controlled route-conflict fairness audit. Values are means over 100 paired seeds. The selected-route columns report the fraction of runs selecting the indicated route class.

Condition	MeshEcho PDR	No-confidence PDR	Δ PDR	Long selected	Short selected	Δ airtime (s)
No extra shadowing	1.000	1.000	0.000	0.000	1.000	0.000
Short branch +5 dB	0.980	0.980	0.000	0.020	0.980	0.000
Short branch +7.5 dB	0.999	0.784	+0.215	1.000	0.880	+15.148
Long branch +7.5 dB	1.000	1.000	0.000	0.000	1.000	0.000

branch is weakened. The 7.5 dB condition is therefore evidence that the candidate-ranking mechanism can detect a short-but-weaker path under this model. It is not evidence of topology-independent superiority. The higher airtime in that condition also shows the cost of selecting the reliable four-hop route instead of the three-hop route. The moderate +5 dB condition is an important negative control: the observed mean delta is zero, so the mechanism does not produce a gain for every small perturbation.

I. Route-Discovery Stress Check

The connected-pair selector was also exercised separately to distinguish multi-hop geometry from route-discovery operability. It selected a shared pool of pairs at graph distance 2–4 using a static link-budget graph whose retained edges had analytical PRR at least 0.90. In a 50-node, 15 km square with 24 pairs, SF7, and 0.5 flows per minute, MeshCore-like installed a route in only 0.063 of route-discovery attempts and achieved 0.049 destination PDR. A 20-seed single-flow check still gave a 0.050 discovery-success rate. The RREP/RREQ transmission ratio was only 0.027 in the low-load run. We therefore treat this configuration as a broadcast route-discovery stress test rather than as evidence that source-route data forwarding is intrinsically inferior. MeshCore-like also emits an RREP immediately upon first destination reception, whereas MeshEcho waits for its candidate-collection window; in a collision-limited flood, this reply-timing asymmetry can affect discovery success. The corresponding raw rows and discovery diagnostics are released separately from the headline comparison.

VI. Discussion

The preliminary results support five observations. First, fixed flooding and fixed source-route caching expose different sides of the LoRa mesh design space. Flooding protects delivery but consumes substantial airtime. Cached source routing saves airtime, but it can underperform when the selected route is fragile. Because the default direct links are almost all high-PRR, the random matrix should be interpreted primarily as a contention-oriented operating point, not as a representative deployment-wide multi-hop calibration.

Second, route confidence is a plausible abstraction, but evidence for its benefit comes primarily from the controlled route-conflict experiment. The random-topology matrix

does not estimate its average benefit reliably because links are mostly high-PRR and cached paths are almost always one hop.

Third, the best operating point depends on traffic. In repeated unicast, route caching is already strong and MeshEcho mainly improves efficiency. In mixed traffic, its reliability gain over source-route caching includes bounded recovery unavailable to the fixed baselines. The current-code sensitivity audit estimates a small but nonzero mean effect from route-miss fallback ($+0.009 \pm 0.018$ ACK PDR), with a corresponding airtime cost. This remains a protocol-bundle result, not an isolated estimate of confidence ranking. Redundancy should therefore be treated as a per-context budget rather than a static network-wide mode.

Fourth, the random-pair probe changes the external-validity interpretation. In the non-saturated multi-hop setting, MeshEcho retains lower airtime than the source-route baseline but does not improve ACK PDR. After disabling additional Smart-CALM timeout retries, the full and no-confidence variants are nearly identical in mean ACK PDR, and their paired interval includes zero. Online adaptation and confidence ranking should therefore not be presented as general-purpose improvements from this probe. The probe is useful evidence against overclaiming, not a replacement for a larger calibrated evaluation.

Fifth, the two experiments answer different questions: route-conflict isolates admission under a known quality difference, while the random matrix measures end-to-end behavior. The main matrix is a common-harness comparison, not a fully budget-matched comparison of all recovery mechanisms; a mechanism study must equalize route-cache lifetime, retry, and fallback budgets as well.

VII. Limitations and Future Work

This work is a packet-level simulation rather than a physical deployment. The model includes airtime, shadowing, probabilistic reception, half-duplex radios, collision, and capture, but not full waveforms, duty-cycle regulation, real firmware queues, or hardware sensitivity variation. The default 50-node, 3000 m, SF9 setting is a contention-oriented, high-PRR regime with almost no multi-hop cached paths. Repeated unicast and the unicast component of mixed traffic use eight fixed pairs. A separate ten-seed random-pair probe is included to test this bias, but it has a harsh, partially disconnected operating point and

is diagnostic rather than a calibrated deployment study. Consequently, the current data are not sufficient to claim calibrated multi-hop deployment performance.

The comparison is fair at the common-harness level but not fully budget-matched: MeshEcho has bounded fallback, while the fixed baselines have no identical recovery controller. The new paired recovery audit shows that disabling route-miss fallback changes the current CALM line by $+0.009 \pm 0.018$ ACK PDR and $+0.023 \pm 0.023$ destination PDR, while reducing airtime by 15.9 s on average. The one-shot random-pair probe disables additional Smart-CALM timeout retries, so the confidence and no-confidence variants have the same timeout-retry budget, but native fallback semantics remain different. The native route-cache TTL is also 300 s for MeshCore-like versus 600 s for MeshEcho; matching it to 600 s raises MeshCore-like ACK PDR by $+0.138 \pm 0.031$ in the paired audit and reduces the remaining matched gap. The separate two-retry Smart-CALM run is a recovery evaluation and is not used as evidence for a confidence-only gain. Moreover, reception, forwarding jitter, and exploration share one stream in the frozen matrix: same seeds fix topology, traffic, and static shadowing, but not strict event-by-event common random numbers. The released independent-stream option removes this particular coupling for new reruns, but does not solve the different-transmission-count problem. All packet kinds also use the same configured payload length, so source-route header growth and control-frame format differences are omitted from the airtime estimate. The connected-pair stress check also contains a route-reply timing difference: MeshCore-like replies immediately, while MeshEcho waits for candidate collection. This is another reason that the stress check is not used as a neutral cross-protocol headline comparison.

The historical Smart-CALM ablation CSVs predate the no-fallback threshold fix included in the released implementation and are not used for the main quantitative claims. They remain in the repository only to preserve version traceability; a fresh ablation rerun is required before interpreting those variant rows.

The route-conflict ablation uses one seven-node topology, deterministic permitted edges, zero stochastic shadowing, staggered route-request timing, and a static channel. Its 100 seeds are repeated stochastic trials on that topology, not 100 independent network layouts. The confidence estimate and subsequent forwarding use the same static link realization, so stale or noisy confidence is not tested. Finally, the comparison does not include a separately implemented ETX or airtime-aware routing baseline.

Future work will expand the random-pair probe into a pre-registered full matrix, calibrate a genuine multi-hop regime using direct-link PRR, route-discovery success, and cached-route hop distributions, equalize retry and fallback budgets, use the independent channel-random stream (or keyed per-event draws), model variable packet headers,

introduce time-varying links, and validate the design with hardware-in-the-loop experiments.

VIII. Conclusion

This paper presented a preliminary evaluation of MeshEcho confidence-aware routing for airtime-constrained LoRa mesh IoT networks. Using a reproducible packet-level simulator, we compared managed flooding, source-route caching, and the calm-mesh MeshEcho prototype. The results show that fixed routing modes have clear operating boundaries: flooding is reliable but airtime-intensive, while cached source routing is efficient but can lose unicast reliability in mixed traffic. In the frozen 50-node mixed matrix, MeshEcho reaches 0.735 ACK-confirmed unicast PDR with 840 s of airtime, compared with 0.389 PDR and 971 s for source-route caching. The controlled audit further shows that the route-ranking gain disappears under null, moderate, and reverse controls, which bounds the mechanism claim to conditions where link-quality evidence is informative. The one-shot random-pair probe confirms the same boundary: MeshEcho keeps an airtime advantage but does not show a PDR advantage over source-route caching, and Smart-CALM is nearly identical to its no-confidence control. Because fixed pairs, limited probe seeds, route-discovery stress cases, and unequal native recovery semantics remain, these results should be read as preliminary common-harness evidence rather than a universal ranking of LoRa mesh protocols.

Reproducibility and Declarations

The MeshEcho simulator, routing implementation, experiment scripts, raw results, and paper source are released in the accompanying repository: <https://github.com/BH4ME/lora-mesh-routing-sim> (repository release 2.1.12). No human participants or personal data were involved. The authors report no conflicts of interest and no external funding. Author contributions cover protocol design, implementation, experiment design, analysis, and manuscript preparation. AI-based tools assisted with code inspection and language editing; the authors verified the implementation, numerical results, and final manuscript.

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